

Q&A Preparation — NIR-to-RGB Translation

Examiner question bank + technical self-test. Cover the grey lines and answer aloud first.

Part A — Examiner question bank

Tailored to published research areas; treat as educated bets. Tiers: [E]asy, [M]edium, [H]ard, [V]ery hard.

Stefan Vlaski — optimisation, learning theory, distributed/adaptive ML

- [E] What exactly is your training objective, and how was the final checkpoint chosen?
→ *Weighted L1 + MS-SSIM + light PatchGAN + DINOv2/CLIP/ResNet feature ensemble; checkpoint by best validation CLIP cosine.*
- [M] Batch size 1 — what does that do to gradient noise/stability, and why acceptable?
→ *Memory-bound at 256^2 with 116M + 3 frozen extractors; zero-init residual scales and a strong L1 anchor stabilise it.*
- [H] Your loss has 6–7 terms. How are weights set, and did you check for conflicting gradients between the feature terms (negative transfer)?
→ *Hand-tuned with warm-up; ablations are single runs, so component conclusions are exploratory, not confirmatory — say so.*
- [H] Why does feature-level KD from the teacher beat matching ground truth alone?
→ *It matches the teacher’s intermediate representations, preserving the feature behaviour the downstream models read, not just pixels.*
- [V] Matching $F(T(x)) \approx F(\text{RGB})$ is necessary, not sufficient, for task recovery. Argue the link.
→ *Covariate-shift view: downstream head is fixed; matched input features $\Rightarrow$ matched outputs — but features are pooled/partial, hence “agreement, not accuracy.”*
- [V] Could this be made online/adaptive on-device, and what would break?
→ *Possible in principle; determinism (F3), calibration drift and stability of feature targets would all be at risk.*

Cong Ling — information theory, coding/lattices, ML theory

- [E] One-sentence contribution; why is NIR $\rightarrow$ RGB intrinsically hard?
→ *Behaviour-centred translator; hard because colour is absent (one-to-many) and must be inferred from material/texture/context.*
- [M] How much colour is genuinely recoverable from NIR, and where does it come from?
→ *Only what data priors allow — material/texture/scene context; not from the band itself.*
- [H] A deterministic L1/feature model tends to the conditional mean $E[\text{RGB} \mid \text{NIR}]$, which should blur. Why isn’t it blurred?
→ *MS-SSIM + GAN + feature terms pull off the pixel-mean; selection on CLIP not PSNR; you deliberately trade PSNR for feature agreement.*
- [H] INT8 PTQ is scalar uniform quantisation. Would non-uniform/vector/lattice quantisation help at the same bitrate? Why uniform?
→ *Likely yes in theory, but edge kernels (XNNPACK/Hailo) only support uniform-INT8 — a hardware constraint, not a theoretical preference.*
- [V] Is there an information-theoretic ceiling on gap closure given the MI between NIR and RGB? How would you estimate it?
→ *Conceptually yes; could bound via mutual information / rate-distortion, but estimating MI on this data is itself hard — be honest it’s untested.*

Bruno Clerckx — wireless comms, signal processing, real systems

- [E] Sensor setup, resolution, frame rate?
→ *Two Pi Camera Module 3 sensors via beam-splitter, one behind an 850 nm filter; 256×256 stored; capture every 5 s.*
- [M] You report model-inference latency — what about end-to-end capture $\rightarrow$ output?
→ *Not separately timed; a known, stated limitation. Full pipeline delay untested.*
- [H] The beam-splitter halves incoming light, but NIR’s value is low light. SNR implications? Does it work at night?
→ *Concede: dataset is daytime passive NIR; low-light/active-illumination not characterised — the operating envelope (blur/noise) bounds this.*
- [H] The three NIR channels are highly correlated (single band). Isn’t 3-channel input wasteful?
→ *Yes — a 1-channel input could shrink the intro conv; kept 3-ch for drop-in compatibility with RGB-shaped pipelines.*
- [V] Real night deployment uses active NIR illumination, whose statistics differ from your daytime data. Does the translator survive that domain gap?
→ *Untested; likely needs domain-matched data. This is the deepest systems critique — answer honestly, link to the robustness envelope.*

Cross-cutting (any examiner)

[M] Pseudo-labels — isn't evaluating against the model's own RGB output circular?

→ *Independent probes (DINOv2-L, SigLIP2) held out of training still close > 0.5.*

[M] Why is detection closure so much weaker than segmentation?

→ *Detection leans on small objects + fine texture, reproduced least faithfully; seg/depth lean on large-scale structure.*

Part B — Technical self-test (do you understand the building blocks?)

NAFNet / architecture

[E] What is SimpleGate?

→ *Split channels in half, multiply the two halves element-wise.*

[M] "Activation-free" — so is it linear? Where's the non-linearity?

→ *Not linear: the SimpleGate multiply (product of two linear maps) and channel attention are nonlinear. "Activation-free" = no fixed pointwise ReLU/GELU.*

[H] Why does that help INT8 specifically?

→ *No saturating activations ⇒ predictable ranges, nothing to clip-calibrate; plus a small, standard op set.*

[H] Why disable NAFNet's global input→output residual?

→ *It would force colour to be a delta on NIR; wrong inductive bias across modalities.*

Knowledge distillation

[M] Why can a 1.7M student mimic a 116M teacher?

→ *It learns an already-solved, smoother target (teacher output) rather than the task from scratch — "dark knowledge".*

[H] Feature-KD vs output-KD?

→ *Output-KD matches pixels; feature-KD matches intermediate frozen-feature activations between student and teacher.*

Quantisation

[E] What does INT8 store?

→ *real $\approx$ scale $\cdot$ (q - zero-point); 256 levels.*

[M] PTQ vs QAT?

→ *PTQ quantises after training via calibration ranges (no retraining); QAT simulates rounding during training so weights adapt — usually more accurate.*

[M] What does calibration do?

→ *Runs representative data through FP32, collects activation MinMax to set per-tensor scales; weights per-channel.*

[H] Why is INT8 faster than FP32 on CPU, not just smaller?

→ *Higher integer SIMD throughput, lower memory bandwidth/cache pressure, narrower kernel set.*

[H] Why "mixed" INT8 not full INT8?

→ *PixelShuffle=DepthToSpace has no INT8 kernel at opset 17 (and LiteRT LayerNorm reductions); those stay FP32, feeding convs excluded so they aren't stranded between quant/dequant.*

Losses

[E] Why L1 not L2 for the pixel term?

→ *L1 blurs less; L2 averages plausible colours.*

[M] MS-SSIM and why include it?

→ *Multi-scale structural similarity (luminance/contrast/structure across scales); structural fidelity beyond per-pixel error.*

[M] PatchGAN + LSGAN — what does each give?

→ *PatchGAN judges 70×70 patches ⇒ local texture realism; LSGAN uses least-squares ⇒ stabler gradients.*

[H] What is a perceptual loss, and why drop the standard VGG one?

→ *L2 in a pretrained net's feature space; replaced single VGG space with a DINOv2/CLIP/ResNet ensemble because the output is consumed by those spaces.*

Alignment / geometry

[M] When is a single homography valid for a whole scene?

→ *Pure rotation about a common centre, planar scene, or no parallax. Beam-splitter gives a shared optical centre ⇒ valid at all depths.*

[H] Why does a side-by-side rig break that?

→ *Baseline ⇒ depth-dependent parallax (disparity $\propto 1/\text{depth}$); no single 2D warp can undo it.*

[M] SIFT, Lowe ratio, RANSAC — one line each?

→ *SIFT: scale/rotation-invariant keypoints+descriptors. Lowe: keep match if nearest/second < 0.75. RANSAC: fit homography from random 4-point sets, keep most-inlier model.*

Metrics

[M] mAP@0.5 vs mAP@0.5:0.95?

→ *AP = area under precision-recall per class, averaged. @0.5 = one IoU threshold; @0.5:0.95 = averaged over thresholds.*

[M] Dice vs IoU?

→ *Dice = $2|A \cap B|/(|A| + |B|)$ (= F1); IoU = $|A \cap B|/|A \cup B|$; monotonically related.*

[H] What does CLIP-FID capture that per-image cosine doesn't?

→ *A distribution-level distance (Fréchet distance between feature-set Gaussians) — whether the set of outputs matches, not just per-image.*

Gap-closure metric

[M] Write it; explain the denominator.

→ *$(m_{\text{trans}} - m_{\text{NIR}})/(m_{\text{RGB}} - m_{\text{NIR}})$; denominator is total NIR→RGB headroom, normalising improvement against the max recoverable.*

[H] For embeddings, what is m_{RGB} , and for depth (lower=better) how is the sign handled?

→ *$m_{\text{RGB}} = 1$ (RGB self-cosine); for lower-is-better the harness inverts the difference so positive still means more RGB-like.*

Export stack

[H] Why three operator fixes for Hailo?

→ *Restore explicit conv kernel shapes; rewrite SimpleGate as two 1×1 identity-selector convs into the multiply; decompose global-average-pool into a two-stage reduction. All numerically exact.*

Part C — General / panel question bank

Broad-viva questions any non-specialist marker might ask: framing, process, judgement, impact.

Framing & motivation

- [E] In one sentence, what problem does this solve and for whom?
→ *An engineer with a working RGB perception stack who needs to run it on NIR without retraining.*
- [E] Why not just collect NIR data and retrain?
→ *Retraining needs open weights + labelled NIR per task; one translator reuses every frozen model unchanged.*
- [M] The single most important result?
→ *Pixel fidelity doesn't predict downstream usefulness — Pix2Pix ties on PSNR/SSIM yet recovers nothing downstream.*
- [M] Who would deploy this, and what do they still need that you haven't provided?
→ *Night/biometric/inspection NIR users; they'd need person-class validation, active-illumination data, and end-to-end timing.*

Process & judgement

- [M] Hardest part technically?
→ *A clean parallax-free paired dataset, and a quantise/export path that survives onto the Pi.*
- [M] What did you try that didn't work?
→ *NIR-gradient-weighted L1 barely helped; onnx2tf broke LayerNorm layout (moved to litert-torch).*
- [H] What would you change if restarting?
→ *QAT instead of PTQ; collect person/low-light data under ethics; budget end-to-end timing from the start.*
- [H] Effort that wasn't worth it in hindsight?
→ *The pixel-side loss experiments — the win was entirely on the feature side.*

Results interpretation (watch for over-claiming)

- [M] Does gap closure 0.59 mean the model is “59% accurate”?
→ *No — it's 59% of the recoverable behavioural agreement with the RGB model, not task accuracy.*
- [H] Detection is +0.08, segmentation +0.59. Is detection really a success?
→ *Positive but weak — honestly the weakest result; small-object/fine-texture reason.*
- [H] Could any gains be an evaluation artefact rather than real improvement?
→ *Possibly: pseudo-labels + partly non-independent probes; mitigated by held-out DINOv2-L/SigLIP2, stated as a limitation.*

Impact, ethics, legality

- [E] Privacy measures in the dataset?
→ *Rig tilted up, 256×256 low-res, dataset unpublished — no identifiable imagery retained.*
- [M] This enables better NIR surveillance — dual-use concern?
→ *Acknowledge it; note privacy-by-design choices and that person-tracking is out of scope/untested.*
- [M] Licensing constraints?
→ *YOLOv8 is AGPL — used only as an offline eval probe, not deployed; the student is original work.*

Positioning & scalability

- [M] How is this different from the closest prior work (Pix2Next)?
→ *Reverse direction (NIR→RGB), ensemble used purely as a training signal on an unmodified backbone, behaviour-centred evaluation across a task panel.*
- [H] Does it scale to higher resolution / video?
→ *Untested above 256^2 ; video needs temporal consistency (flagged as further work — flicker risk).*
- [H] Cheapest way to reproduce/extend?
→ *Rig is a Pi + two commodity cameras + a beam-splitter; reusable for other cross-spectral pairings.*

Viva classics — have crisp answers ready

- [E] Your contribution versus off-the-shelf components?
→ *The combination + evaluation standard: restoration backbone for cross-spectral synthesis, foundation-ensemble supervision, the aligned-capture rig, behaviour-centred gap-closure.*
- [H] Drop one component — which, and what's lost?
→ *Drop the GAN term (smallest effect, mainly minor detection help); keep the feature ensemble (No-FM ablation shows it's essential).*
- [V] Is any of this novel, or just integration?
→ *Parts are published; novelty is (a) NAFNet for NIR→RGB (not previously reported), (b) foundation-ensemble as the training signal, (c) behaviour-centred annotation-free evaluation across a task panel.*
- [V] Weakest claim in the thesis — defend or retract?
→ *Name one (e.g. detection usefulness, or generality beyond daytime urban) and defend it as bounded, not universal.*

Part D — Deep technical probes

Second-order questions that separate “I used it” from “I understand it.” Mostly [H]/[V].

Quantisation, deeper

[H] Symmetric vs asymmetric — which for weights, which for activations, and why?

→ *Weights: symmetric, per-channel (zero-point 0, centred). Activations: asymmetric (zero-point $\neq 0$) to fit shifted/non-negative ranges; Hailo I/O is UINT8.*

[V] Why does depth degrade most under INT8 (largest relative loss)?

→ *Depth depends on subtle gradients/relative ordering; quantisation injects high-frequency error that perturbs the ordering, and per-image min-max-normalised L1 amplifies small absolute changes.*

[H] You calibrate on 100 training frames — what bias does that risk?

→ *If calibration $\neq$ deployment distribution, activation ranges are mis-set $\Rightarrow$ clipping/saturation. Representative but daytime-only — a stated limitation.*

[H] Why must LayerNorm stay FP32 under quantisation?

→ *Its mean/variance reductions need precision; integer variance is unstable and there’s no good INT8 kernel — so it’s left in FP32.*

[V] How do you actually guarantee bit-identical 8-bit outputs (determinism, F3)?

→ *INT8 integer kernels are exact/deterministic; fixed runtime + thread config; verified zero mismatched pixels on repeated runs. FP32 may vary across hardware — tested on the fixed runtime.*

Feature-matching mechanics

[H] Do you match spatial feature maps or pooled embeddings, and at which depth?

→ *Training matches intermediate feature-map activations (early/mid) per backbone — spatial, so localisation is preserved; pooled cosine is only for evaluation.*

[M] Why compute feature losses at 224×224 when training at 256^2 ?

→ *The frozen backbones expect ~ 224 input; resize to their native resolution before extracting.*

[H] Gradients and frozen networks — if the extractors are frozen, how do they train the generator?

→ *Frozen = no weight update, but gradients still flow through them to the generator; they must stay differentiable and FP32 for stable backprop.*

[V] Why an ensemble rather than one strong backbone — and what’s the failure mode?

→ *Different backbones encode different statistics (self-sup, vision-language, supervised CNN); risk is one term dominating or conflicting gradients — managed by weighting (DINOv2 40 vs ResNet 5) and warm-up.*

Adversarial & architecture

[H] Why does LSGAN reduce vanishing gradients vs a vanilla (BCE) GAN?

→ *Sigmoid+BCE saturates for confident samples; least-squares penalises by distance from the boundary, so gradients persist (it minimises a Pearson χ^2 divergence).*

[H] A 70×70 PatchGAN — what can it not enforce, and why does that matter here?

→ *It judges local patches, so it can’t enforce global colour consistency — a known $\text{pix}2\text{pix}$ weakness; NAFNet’s deep bottleneck (large receptive field) compensates.*

[H] Why is BatchNorm-free (LayerNorm) important at batch size 1?

→ *BatchNorm statistics are unreliable for tiny batches; LayerNorm is per-sample, so training is stable — $\text{pix}2\text{pix}$ ’s BatchNorm-heavy U-Net is a liability here.*

Alignment, deeper

[V] SIFT across the NIR/RGB gap: foliage inverts in brightness — why does matching still work?

→ *SIFT uses gradient-orientation histograms, robust to monotonic intensity change but not contrast inversion; matches concentrate on structural edges (buildings), which is why vegetation yields fewer inliers.*

[H] When can RANSAC return a bad homography even with many inliers?

→ *Near-degenerate point configs (dominant single plane, low texture, collinear points); mitigated by an inlier-count threshold and discarding degenerate fits.*

[H] What exactly does DepthToSpace/PixelShuffle do, and why no INT8 kernel?

→ *Rearranges $(C r^2, H, W) \rightarrow (C, Hr, Wr)$ (sub-pixel up-sample); just a reshuffle, but the runtime lacked an INT8 implementation at opset 17, so it (and feeding convs) stays FP32.*

Metrics & evaluation, deeper

[V] Using the model’s own RGB detections as “ground truth” — what bias does that bake in?

→ *It measures agreement with the RGB model’s behaviour including its mistakes, not correctness; native-RGB self-agreement is 1 by construction.*

[H] The Dice 5% pixel threshold — what bias does it introduce?

→ *Excludes tiny/absent classes to avoid noisy Dice, but only evaluates where a class is prominent — a selection bias that inflates scores.*

[H] Gap closure: ratio of aggregate means, or mean of per-image ratios? Why does it matter?

→ *Computed on aggregate metrics (e.g. class-mean Dice); ratio-of-means $\neq$ mean-of-ratios, and per-image denominators can be near-zero — the harness handles the zero-denominator case explicitly.*

Drill these three first — most likely to expose a shaky answer: where the nonlinearity is if it’s “activation-free”; why INT8 is faster (not just smaller); when a single homography is valid. If you don’t know: say so, then reason aloud — honesty is marked.